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# **Tool Lists for Hospital Fixed Asset Tracking Sensor**

Imperial Healthtech

***Edited by:***  
*Farros Ramzy*

## Tools Needed for The Project

To start the Asset Tracking Sensor System project, some tools are needed so the maintainers can build the product during production.

The main idea of this project is to detect the final position of a fixed item in the hospital from class C and above, to class A. This means the system must be suitable for large hospitals that handle at least 100 in-bed patients per day.

Since the project requires a wireless product to reduce manual effort for users, wireless tools such as the ESP32 need to be maintained for the project. And other than that, sensors that detect items, such as UHF RFID tags, also need to be maintained.

And in case of another solution for the product of this project, Bluetooth Low Energy (BLE) module & tags can also be used instead of the UHF RFID system.

Below are the lists of items that can be bought as tools for this project, with their descriptions and price estimates.

### 1. ESP32 Microcontroller Device

The items below are candidates for the ESP32 Microcontroller, along with their prices.

| No. | Supplier     | Description                                                                                                                                                                                                                                                        | Estimated Price per Product                                     |
|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1.  | DigPart      | ESP32 WROOM 32 4MB with Wifi and Bluetooth for IoT Development Board, ESP32S CP2102 Board.<br>Connect with micro-USB.<br><br>Link to the shop site:<br><a href="https://shopee.co.id/imamsholahudin/2187290352">https://shopee.co.id/imamsholahudin/2187290352</a> | Rp. 74.000<br><br>If need 4, total price:<br><b>Rp. 296.000</b> |
| 2.  | solarperfect | ESP 32 WROOM 32 with WiFi + BLE Development Board.<br>Connect with micro-USB.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunCnnnL/">https://tk.tokopedia.com/ZSunCnnnL/</a>                                                               | Rp. 69.700<br><br>If need 4, total price:<br><b>Rp. 278.800</b> |
| 3.  | Kyware       | ESP32 ESP 32 WROOM WIFI Bluetooth, CP2102 CH340 with 38 Pins + BLE IOT Development Board.<br>Connect with USB type C.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunXmqSY/">https://tk.tokopedia.com/ZSunXmqSY/</a>                       | Rp. 68.090<br><br>If need 4, total price:<br><b>Rp. 272.360</b> |
| 4.  | Supreme ID   | ESP32 ESP-32 WROOM WIFI + BLE IoT Development Board (38Pin) CH9102X.<br>Connect with micro-USB.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSun4eW4s/">https://tk.tokopedia.com/ZSun4eW4s/</a>                                             | Rp. 74.500<br><br>If need 4, total price:<br><b>Rp. 298.000</b> |

By reviewing the item descriptions from each shop, the ESP32 product can be elaborated on the table below.

| No. | Product                                    | Key Features          | Pros                                                                                                                               | Cons                                                                                                                  | Verdict                             |
|-----|--------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 1.  | DigPart ESP32 WROOM 32 (CP2102, Micro-USB) | WiFi + BLE, micro-USB | <ul style="list-style-type: none"> <li>Reliable standard ESP32</li> <li>Easy USB programming</li> <li>Widely compatible</li> </ul> | <ul style="list-style-type: none"> <li>Micro-USB (weaker)</li> <li>Slightly higher price</li> </ul>                   | Safe choice but not best value      |
| 2.  | solarperfect ESP32 WROOM 32 (Micro-USB)    | WiFi + BLE, micro-USB | <ul style="list-style-type: none"> <li>Cheapest among mid-range</li> <li>Good for prototyping</li> </ul>                           | <ul style="list-style-type: none"> <li>Older connector</li> <li>Unknown board quality</li> </ul>                      | Budget option                       |
| 3.  | Kyware ESP32 WROOM (USB-C)                 | WiFi + BLE, USB-C     | <ul style="list-style-type: none"> <li><b>USB-C (stronger, modern)</b></li> <li>Lowest price</li> <li>Good convenience</li> </ul>  | <ul style="list-style-type: none"> <li>Same power consumption issue</li> <li>Quality depends on the seller</li> </ul> | <b>Best overall pick</b>            |
| 4.  | Supreme ID ESP32 (CH9102X, Micro-USB)      | WiFi + BLE, micro-USB | <ul style="list-style-type: none"> <li>Stable chip (CH9102X)</li> <li>Reliable dev board</li> </ul>                                | <ul style="list-style-type: none"> <li>Highest price</li> <li>Still micro-USB</li> </ul>                              | Overpriced for the same performance |

And to compare each of the products, the ESPs above can be listed in the checklist table as follows:

| No. | Product            | USB-C | Lowest Price | Reliable Chip | Has BLE | Good for Proto-type | Notes               |
|-----|--------------------|-------|--------------|---------------|---------|---------------------|---------------------|
| 1.  | DigPart ESP32      | x     | x            | ✓             | ✓       | ✓                   | Older micro-USB     |
| 2.  | solarperfect ESP32 | x     | ! (mid)      | ! (uncertain) | ✓       | ✓                   | Quality unclear     |
| 3.  | Kyware ESP32       | ✓     | ✓            | ! (uncertain) | ✓       | ✓                   | Best overall        |
| 4.  | Supreme ID ESP32   | x     | x            | ✓             | ✓       | ✓                   | Slightly overpriced |

From the table above, all these ESP32 microcontrollers are equipped with Wi-Fi and BLE, which is great for a quick prototype of the receiver or gateway for the project.

However, most of these ESPs are too large and power-hungry for a final battery tag. The board quality also varies, and they use micro-USB versions, which are older/mechanically weaker than USB-C. These microcontrollers have similar functions, so differences are mostly the connector, USB chip, and seller quality, not core capability.

The strongest advantage for practice is the ESP32 from the Kyware supplier, since it uses the USB-C connector and has the lowest listed price among the four, making it the best convenience/value option for prototyping.

## 2. UHF RFID Module

These UHF RFID modules in the table below are mostly the same. They are basically using the same standard UHF tag detection, which is ISO 18000-6C (the commonly used in hospitals and industrial sectors in Indonesia). The only differences are the price, signal strength features, the connector cable or wire type, and the market from which to buy these items.

The options for the production tool are listed in the table below:

| No. | Supplier                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                               | Estimated Price per Product                                               |
|-----|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1.  | Shenzhen Jingchen Automation Co. Ltd. | UHF RFID ESP32 module with 0-5dbi antenna.<br>Connect with jumper wires to USB connector.<br><br>Link to the shop site:<br><a href="https://indonesian.alibaba.com/product-detail/Integrated-UHF-RFID-Module-ESP32-for-1601576007388.html">https://indonesian.alibaba.com/product-detail/Integrated-UHF-RFID-Module-ESP32-for-1601576007388.html</a>                                                                                                      | Rp. 984.623<br><br>If need 4, total price:<br><b>Rp. 3.938.492</b>        |
| 2.  | Rememberr shopp                       | RFID UHF Radio Frequency Identification UHF Reader Support 840-960MHz Air Recognition.<br>Connect with jumper wires to Breadboard.<br><br>Link to the shop site:<br><a href="https://shopee.co.id/rememberrshopp/48153426755">https://shopee.co.id/rememberrshopp/48153426755</a>                                                                                                                                                                         | Rp. 2.992.996<br><br>If need 4, total price:<br><b>Rp. 11.971.984</b>     |
| 3.  | TOKO SRCrohimah SERJANAAA14           | YRM100 Module UHF RFID reader Module USB Writer TTL 1-5.5dbi Antenna.<br>Connect with jumper wires to Breadboard, or jumper wires to USB connector.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunqCHx9/">https://tk.tokopedia.com/ZSunqCHx9/</a>                                                                                                                                                                                | Rp. 3.250.000<br><br>If need 4, total price:<br><b>Rp. 13.000.000</b>     |
| 4.  | IFAST (rfid-life)                     | UHF RFID Module 3 meters with TTL to USB connection.<br><br>Only order online via email.<br><br>Link to the shop site:<br><a href="https://rfid-life.com/product/RFID-reader-UHF-module-uhf-card-reader-identification-distance-3-meters-TTL-to-USB-interface-915mhz-European-standard.html">https://rfid-life.com/product/RFID-reader-UHF-module-uhf-card-reader-identification-distance-3-meters-TTL-to-USB-interface-915mhz-European-standard.html</a> | € 33.19<br><br>If need 4, total price:<br><b>€ 132.76 (Rp. 2.588.820)</b> |

By reviewing the item descriptions from each shop above, the UHF RFID product can be elaborated on the table below.

| No. | Product                              | Key Features                  | Pros                                                                                                                                      | Cons                                                                                                                  | Verdict                         |
|-----|--------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 1.  | Shenzhen Jingchen (ESP32 integrated) | ESP32 + RFID, 0–5 dBi antenna | <ul style="list-style-type: none"> <li>• <b>All-in-one (ESP32 + RFID)</b></li> <li>• Lowest cost</li> <li>• Easier integration</li> </ul> | <ul style="list-style-type: none"> <li>• Limited antenna gain</li> <li>• DIY wiring needed</li> </ul>                 | <b>Best value + integration</b> |
| 2.  | Rememberr shopp UHF Reader           | 840–960 MHz reader            | <ul style="list-style-type: none"> <li>• Strong industrial support</li> <li>• Wide compatibility</li> </ul>                               | <ul style="list-style-type: none"> <li>• Very expensive</li> <li>• External integration needed</li> </ul>             | Overkill for a prototype        |
| 3.  | YRM100 UHF Module                    | TTL + USB, 1–5.5 dBi antenna  | <ul style="list-style-type: none"> <li>• Flexible connection</li> <li>• Good antenna options</li> </ul>                                   | <ul style="list-style-type: none"> <li>• Expensive</li> <li>• Complex setup</li> </ul>                                | Mid-high tier option            |
| 4.  | IFAST RFID (3m range)                | TTL to USB, ~3m read          | <ul style="list-style-type: none"> <li>• Defined read range</li> <li>• Cheapest (EU pricing)</li> </ul>                                   | <ul style="list-style-type: none"> <li>• Limited purchase access (email)</li> <li>• Possibly lower support</li> </ul> | Good budget alternative         |

And to compare each of the products, the ESPs above can be listed in the checklist table as follows:

| No. | Product           | Cheap | ESP32 Built-in | Easy Setup    | Long Range    | Good for Proto-type | Notes            |
|-----|-------------------|-------|----------------|---------------|---------------|---------------------|------------------|
| 1.  | Shenzhen Jingchen | ✓     | ✓              | ! (uncertain) | ! (uncertain) | ✓                   | Best value       |
| 2.  | Rememberr shopp   | x     | x              | ! (uncertain) | ✓             | x                   | Very expensive   |
| 3.  | YRM100 Module     | x     | x              | ! (uncertain) | ✓             | ! (uncertain)       | Complex setup    |
| 4.  | IFAST RFID        | ✓     | x              | ! (uncertain) | ! (uncertain) | ✓                   | Hard to purchase |

From the comparisons above, it can be concluded that the best product is from the Shenzhen Jingchen shop, since it states that the sensor is already integrated with ESP32 when the others are not.

These sensors can certainly detect most UHF RFID tags that do not use unusual protocols, since they use a common UHF standard, which is good for checkpoint or gate detection. And the RFID tags also need no battery when they are embedded in the fixed item for detection, which makes them good for maintenance.

However, it is known that the RFID sensor cannot determine the exact location of the tagged item and is only useful for indicating the item's status when it passes the sensor. (Not possible for real-time tracking.) With this, this type of sensor is better suited for access control and checkpoints than for tracking whether the item is inside or outside the room.

### 3. UHF RFID Tags (ISO 18000-6C Standard)

If UHF RFID will be used for this project, this RFID tag below can be used.

| No. | Supplier        | Description                                                                                                                                                                                                             | Estimated Price per Product                                                               |
|-----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1.  | Electron ID_NEW | EL-UHF-AD-238U8-CPL UCODE 8 RFID Label Tag Coated Paper Sticker EPC.<br>Frequency: 860 – 960 MHz<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunbTVNK/">https://tk.tokopedia.com/ZSunbTVNK/</a> | Rp. 1.750<br><br>Need to discuss total price with the seller if need to buy it as a roll. |

### 4. BLE (Bluetooth Low Energy) Module / Tags

As alternatives for a different approach to the project goal, the items below are candidates for the Bluetooth Low Energy module, along with their prices.

| No. | Supplier         | Description                                                                                                                                                                                                                    | Estimated Price per Product                                      |
|-----|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1.  | Rahyastore       | NRF52810 Bluetooth Module BLE 4.2 Low Power Bluetooth.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunpgghNQ/">https://tk.tokopedia.com/ZSunpgghNQ/</a>                                                | Rp. 250.000<br><br>If need 2, total price:<br><b>Rp. 500.000</b> |
| 2.  | Kunti Rasa Randa | NRF51822 Bluetooth beacon module iBeacon base station UUID applet shakes the wireless Bluetooth device.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSuns2x4s/">https://tk.tokopedia.com/ZSuns2x4s/</a> | Rp. 237.000<br><br>If need 2, total price:<br><b>Rp. 474.000</b> |
| 3.  | Fortuna Jogja    | NRF51822 Low Power Bluetooth 4.0 BLE ARM 2.4 GHz System.<br><br>Link to the shop site:<br><a href="https://tk.tokopedia.com/ZSunshkN9/">https://tk.tokopedia.com/ZSunshkN9/</a>                                                | Rp. 74.000<br><br>If need 2, total price:<br><b>Rp. 148.000</b>  |

By reviewing the BLE descriptions from each shop in the table above, the BLE can be elaborated on in the table below.

| No. | Product                       | Chips    | Pros                                                                                                            | Cons                                                                                 | Verdict                   |
|-----|-------------------------------|----------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|
| 1.  | NRF52810 BLE 4.2 (Rahyastore) | nRF52810 | <ul style="list-style-type: none"> <li>Low power</li> <li>More modern chip</li> <li>Good performance</li> </ul> | <ul style="list-style-type: none"> <li>Higher cost</li> </ul>                        | <b>Best performance</b>   |
| 2.  | NRF51822 iBeacon (Kunti Rasa) | nRF51822 | <ul style="list-style-type: none"> <li>Beacon-ready</li> <li>Easy deployment</li> <li>Stable</li> </ul>         | <ul style="list-style-type: none"> <li>Older chip</li> <li>Medium price</li> </ul>   | Good for quick setup      |
| 3.  | NRF51822 BLE (Fortuna Jogja)  | nRF51822 | <ul style="list-style-type: none"> <li><b>Very cheap</b></li> <li>Low power</li> </ul>                          | <ul style="list-style-type: none"> <li>Fewer features</li> <li>Older tech</li> </ul> | <b>Best budget option</b> |

And to compare each of the products, the BLEs above can be listed in the checklist table as follows:

| No. | Product           | Low Power | Cheap         | Modern Chip | Easy Setup    | Best for Tracking | Notes                   |
|-----|-------------------|-----------|---------------|-------------|---------------|-------------------|-------------------------|
| 1.  | NRF52810          | ✓         | ✗             | ✓           | ✓             | ✓                 | <b>Best performance</b> |
| 2.  | NRF51822 (Beacon) | ✓         | ! (uncertain) | ✗           | ✓             | ✓                 | Easy deployment         |
| 3.  | NRF51822 (Cheap)  | ✓         | ✓             | ✗           | ! (uncertain) | ✓                 | <b>Best budget</b>      |

From the table above, the NRF52810 uses a newer Nordic chip generation that supports Bluetooth 4.2. This means that the sensor can connect to most devices that use a more recent Bluetooth version than the other product.

Since it is designed specifically for low-power IoT applications, it is well-suited for battery-powered asset-tracking tags in hospital environments.

Compared to older BLE chips, it offers better processing capability, improved stability, and more efficient power consumption, which allows the device to run longer on a battery. This module also supports more flexible firmware development, enabling it to be programmed with custom behaviors such as sleep modes, transmission intervals, and filtering logic. However, it comes at a higher cost than the other options and may require slightly more effort to configure.

Overall, it is the best choice when performance, reliability, and long-term deployment are important.

And in comparison, the NRF51822 BLE module from Fortuna Jogja uses the same older chip as the iBeacon version, but it is offered as a more generic and significantly cheaper option.

Unlike the iBeacon module, this version is not preconfigured, so it requires more manual setup and firmware development. This makes it less convenient for beginners but more flexible for users who want to customize the BLE device's behavior. It still offers low power consumption and can function effectively as a BLE tracking tag, but its performance and features are more limited than those of the newer NRF52 series. Its main advantage is cost efficiency, making it a good option for large-scale deployments where many tags are needed, and budget is a primary concern.

## Conclusion & Recommendation

Based on the comparison of the available tools above, the most suitable approach for detecting only the last known position or checkpoint of a tracked hospital asset is a UHF RFID-based system integrated with an ESP32 microcontroller.

UHF RFID technology is specifically designed for checkpoint-based detection, in which tagged items are identified as they pass through a reader. This aligns directly with the system requirement to record the latest detected location rather than continuously track the asset in real time. Additionally, UHF RFID tags do not require batteries, making them cost-effective and low-maintenance for large-scale deployment in hospital environments.

Although BLE modules provide better real-time tracking capabilities, they introduce higher system complexity and power requirements, which are unnecessary for a checkpoint-based system. Therefore, BLE is less efficient for this specific use case.

Overall, UHF RFID offers a simpler, more efficient, and more scalable solution for implementing a last-position tracking system in indoor hospital environments.

However, if the approach is to always detect the last position of the fixed item, BLE is much more useful than UHF RFID, since the tag will trigger a Bluetooth signal to the nearest receiver if it is battery-powered.

And to implement the system effectively using the RFID, the following tool combination is recommended:

1. Core Tracking Technology
  - Use UHF RFID tags (ISO 18000-6C) attached to each asset.
  - Deploy UHF RFID readers/modules at key checkpoints like the room entrances, storage areas, and hallways.
2. Processing & Communication
  - Use an ESP32 microcontroller as the gateway device to:
    - Receive tag data from RFID readers
    - Process checkpoint events
    - Transmit data to the monitoring system via Wi-Fi connection (can be with the MQTT protocol)
3. Module Selection Strategy
  - For rapid prototyping:
    - Use the Shenzhen UHF RFID module (ESP32 integrated) for simplicity.
  - For flexibility and system development:
    - Use separate UHF RFID reader + ESP32 board.
4. System Design Approach
  - Implement an event-based tracking system, where:
    - Each RFID reader logs the tag ID and checkpoint location.
    - The system updates the database with the latest detected checkpoint only.